

Amir R. Isazadeh

Biomedical Engineer • R&D • Computational Biomechanics • Simulation & Medical Device Design

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Professional Summary

3D Computational Designer and R&D Biomedical Engineer specializing in **procedural 3D modeling**, **biomedical product development**, and **automated pipelines** for custom-fit anatomical designs.

Expert in end-to-end automated, **Python-driven** geometric processing: from complex 3D scans (point clouds/meshes) to **ergonomic, manufacturable, and clinically viable medical designs**.

Proven ability to **accelerate scan-to-model** preprocessing by $\approx 95\%$ and boost **simulation performance 100×** through scalable, parametric workflows optimized for **manufacturing**.

Strategic cross-functional partner **bridging engineering, design, and clinical teams** to translate complex biomedical requirements into **high-performance, tailored products**.

Experience

R&D Biomedical Engineer: 3D Computational Designer

Alberta, Canada

Head and Neck Simulation Center, University of Alberta

2023–2025

- **First-of-its-kind Automated Design Pipeline:** Engineered an end-to-end parametric pipeline processing MRI/CT scans → point clouds → meshes → patient-specific surgical designs. Open-sourced as VSP-BioSim and published in PLOS ONE.
- **Robustness & Automation at Scale:** Engineered a zero-touch Python pipeline to generate individualized designs; orchestrated 15+ complex geometric and simulation libraries (Open3D, fTetWild, FEBio) to achieve a highly reliable, reproducible workflow. **Validated** via 20-hour non-stop batch processing.
- **100× Performance Boost:** Developed a **real-time FEM solver** (Numba and C++/Cython) for patient-specific plastic surgery planning; 100× faster optimization; **validated**, clinically viable, and **adopted at UofA and Misericordia Hospitals**. Open-sourced as hiFEM + two Q1 publications.
- **Sub-millimeter Precision Surgery:** Designed and prototyped functional, conformal 3D models for custom-fit surgical guiding toolkits; validated designs through **performance testing and feedback loops**; ensured **manufacturability** and alignment with 3D printing constraints. Published in Surgical Oncology.
- **$\approx 95\%$ Reduction in Preprocessing Time:** Slashed MRI-to-simulation workflow from 4 hours to 15 minutes; leveraged **AI-powered image segmentation** with Biomedisa, PyTorch, and scikit-learn; accelerated patient-specific scan-to-model workflows, enabling same-day design and surgery.
- **Cross-Disciplinary Collaboration & Clinical Translation:** Partnered with **surgeons, engineers, and designers** at UofA and Misericordia Hospitals; translated 40+ hours of OR shadowing into clinical 3D design requirements; validated iterative prototypes through performance testing and feedback loops; ensured manufacturability and **surgical ergonomics**; accelerated adoption of design workflows across disciplines.

Computational Design Researcher

Tehran

Center of Excellence in Design, Robotics, and Automation

2021

- Developed a rapid **generative design/optimization** framework; integrated CAD tools (SolidWorks, Inventor) with MATLAB, Python, and FEM solvers; built scalable, **procedural modeling workflows**; enabled upfront design evaluations to **minimize defects**; published as two Q1 papers.
- Led a team of four to pilot the workflow with IKCO (automotive manufacturing); integrated industrial **production constraints** into early-stage CAD modeling; the integration assessment projected an $\approx 35\%$ **reduction in defective parts** by ensuring manufacturability.

Skills & Expertise

- **Geometric Modeling & Processing:** SolidWorks, Fusion 360, Rhino/Grasshopper, Blender (bpy), gmsh, Open3D (Point Clouds), trimesh, shapely, PyVista, meshio, libigl, pymeshlab, MeshMixer, fTetWild, manifold3d, Interactive All-hex Meshing, HexBox, PolyCube.
- **Computational Mechanics & Simulation:** Nonlinear FEA, Soft Tissue Biomechanics, Large Deformations, Visco-hyperelastic Modeling, FEBio, ABAQUS, PolyFEM, Real-time Simulations (NVIDIA Warp, Omniverse, Isaac Sim).

- **Statistics & Data Visualization:** NumPy, SciPy, Matplotlib, Seaborn, ParaView, PyVista, trame-vtk, statsmodels, R, SQL, STATA.
- **AI/ML, Automation & Programming:** Python, PyTorch, scikit-learn, C++, MATLAB, Git, GitHub.
- **Virtual Surgical Planning & Medical Imaging:** AI-Driven MRI Segmentation, Automated Patient-Specific Model Preprocessing Pipeline, 3D Slicer, Biomedisa.
- **Leadership & Clinical Collaboration:** Surgeon & Healthcare Professional Partnerships, Led Multidisciplinary R&D Teams, Mentored Junior Researchers, IEEE Reviewer, MRI Studies Volunteer.
- **Data Engineering:** Scalable data pipelines using SQL (PostgreSQL, SQLite, Supabase, MySQL) or NoSQL (MongoDB, Firebase); FastAPI, REST APIs, Async/Await; Cloud (AWS, Azure), Containerization (Docker).

Education

PhD in Biomedical Engineering

University of Alberta | 2023–2025

- Patient-Specific Virtual Surgical Planning: Biomechanical Simulation & Guided Execution. **GPA: 3.95/4.0**
- Secured a total of **\$270K** in competitive scholarships and research funding; served as a **reviewer for IEEE** and **mentored junior researchers** across cross-disciplinary teams; completed **the PhD in under three years**.

M.Sc. in Mechanical Engineering

Sharif University of Technology | 2019–2021

- Rapid computational optimization of part manufacturing processes. **GPA: 4.0/4.0**

B.Sc. in Mechanical Engineering

Amirkabir University of Technology | 2013–2018

- Parametric design of axial-flow compressors: Procedural modeling. **GPA: 3.9/4.0**

Selected Publications

1. Guided tongue reconstruction: Feasibility analysis of additively manufactured surgical toolkit on a silicone tongue model (2026). *Surgical Oncology*. doi: 10.1016/j.suronc.2026.102384 (**1st author**).
2. Computational analysis of tongue reconstruction surgery: The impact of free flap site and volume on the tongue anatomy and biomechanics (2025). *PLOS ONE*. (under review) (**1st author**).
3. Patient-specific virtual surgical planning for tongue reconstruction: Evaluating hyperelastic inverse FEM with four simulated tongue cancer cases (2025). *Phys Med Biol*. doi: 10.1088/1361-6560/adebd8 (**1st author**).
4. Algorithmically designed flaps in tongue reconstruction: A feasibility analysis (2024). *Int J Comput Assist Radiol Surg*. doi: 10.1007/s11548-024-03062-w (**1st author**).
5. Optimum design of middle-stage tool geometry and addendum surfaces in sheet metal stamping processes using a new isogeometric-based framework (2022). *JEM*. doi: 10.1177/09544054211041046 (**2nd author**).
6. Some improvements on the one-step inverse isogeometric analysis by proposing a multi-step inverse isogeometric methodology in sheet metal stamping processes (2021). *APM*. doi: 10.1016/j.apm.2020.09.032 (**1st author**).

Selected Recognition & Awards

- Graduate Student Support Award (\$5,000) Canada
2024
GPS Awards, University of Alberta
- Nominated for Vanier Canada Graduate Scholarships Canada
2023
CIHR Federal Competition
- Faculty of Rehabilitation Medicine GRAF (\$64,000) Canada
2023
University of Alberta
- Graduate Recruitment Scholarship (\$5,000) Canada
2023
University of Alberta
- Industrial Transformation and Training Centre for Joint Biomechanics Scholarship (\$99,000) Australia
2022
Queensland University of Technology (successfully awarded, declined)
- Higher Degree Research Tuition Fee Sponsorship (\$96,000) Australia
2022
Queensland University of Technology (successfully awarded, declined)
- Recognized as “Outstanding Student” and ranked in the top 5% of graduates Tehran
2021
Sharif University of Technology
- Ranked 11th (out of 13,900) in the national M.Sc. entrance exam Tehran
2018
National Organization of Educational Testing